1. Write down the Bayesian minimum error decision rule for a two-class problem (classes $\mathcal{C}_1$, $\mathcal{C}_2$ and observation vector $\mathbf{x}$) (no need to prove it, just show you know what it is) **[2 credits]**

$$P(C_i|x) = \frac{P(x|C_i) \cdot P(C_i)}{P(x)}$$

Assign x to C_i if $P(C_i|x) > P(C_j|x)$

2. Figure 1 shows the probability density functions $p(\mathbf{x}|\mathcal{C}_1)$ and $p(\mathbf{x}|\mathcal{C}_2)$ respectively. Assume a uniform prior $p(\mathcal{C}_1) = p(\mathcal{C}_2)$.

   (a) Draw the optimal decision boundary in Figure 1 **[2 credit]**

   (b) How would points $x_1, \ldots, x_4$ be classified? **[4 credits]**

a)

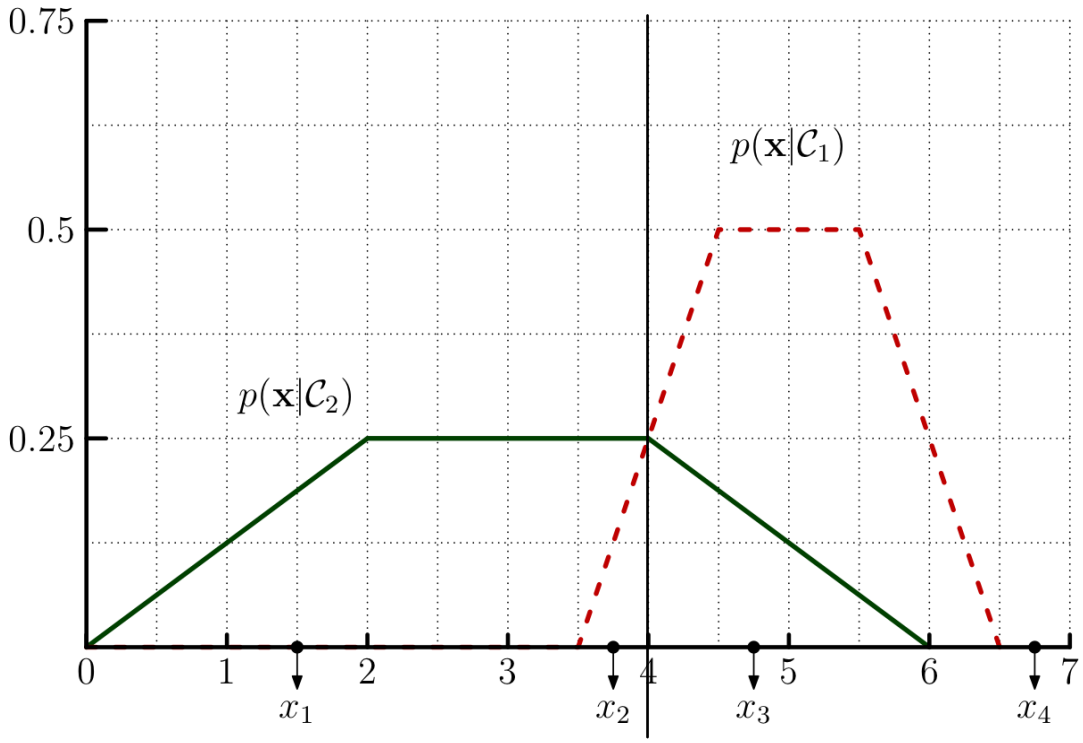


Figure 1: Distributions for the two classes (10 credits)

b) $x_1, x_2 = C_2$

$x_3 = C_1$

x_4 = undefined since both densities are 0

3. Given the decision boundary obtained in 2, there exists a probability $p(\text{error}|\mathcal{C}_i)$ that we obtain a point of class $\mathcal{C}_i$ which is misclassified.

   (a) Which areas in Figure 1 represent the probabilities $p(\text{error}|\mathcal{C}_1)$ and $p(\text{error}|\mathcal{C}_2)$? **[2 credit]**

   (b) For each of the two classes $\mathcal{C}_1$ and $\mathcal{C}_2$, compute the probability $p(\text{error}|\mathcal{C}_i)$ **[1 credit]**

   (c) What is the expected total classification error $p(\text{error})$ given the decision boundary obtained in 2 and prior probabilities $p(\mathcal{C}_1) = 0.8$ and $p(\mathcal{C}_2) = 0.2$? **[2 credit]**

   (d) Let's assume that the cost of misclassification for objects of class $\mathcal{C}_1$ is higher than the cost of misclassification for objects of class $\mathcal{C}_2$: $\lambda_{12} > \lambda_{21}$. Is the optimal decision boundary for minimal classification cost identical to the decision boundary for minimal classification error? If not, is it located to the left or to the right from the optimal classification boundary found in 2? **[1 credit]**

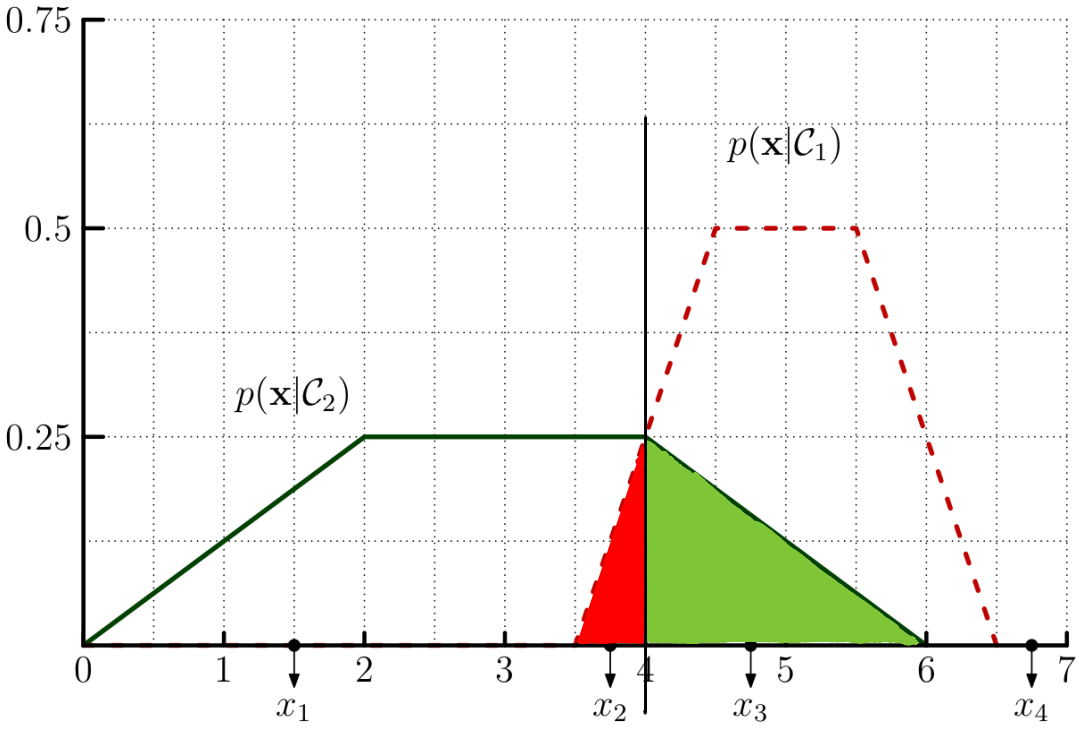


Figure 1: Distributions for the two classes (10 credits)

a) green = $p(\text{error}|C_1)$

red = $p(\text{error}|C_2)$

b) $p(\text{error}|C_1) = \int_{3,5}^{4} p(x|C_1)\,dx = 0,5 \cdot 0,5 \cdot 0,25 = 0,0625$

$p(\text{error}|C_2) = \int_{4}^{6} p(x|C_2)\,dx = \frac{1}{2} \cdot 2 \cdot 0,25 = 0,25$

c) $p(\text{error}) = p(\text{error}|C_1) \cdot p(C_1) + p(\text{error}|C_2) \cdot p(C_2) =$

$0,0625 \cdot 0,8 + 0,25 \cdot 0,2 = 0,05 + 0,05 = 0,1$

d) The boundary is not the same. Since misclassifying C_1 is more expensive, we want to minimize $p(\text{error} | C_2)$ so we move the decision boundary to the left.

Pet	x_1	x_2	x_3
	0	1	0
	1	1	1
	0	0	1
	0	1	0
	0	1	1
	1	1	0
	1	1	1
	0	0	0
	0	1	1

Obstacle	x_1	x_2	x_3
	0	1	0
	1	0	1
	0	0	1
	0	1	1
	1	1	0
	1	0	0

Exercise 2.1 Questions

1. Which feature is best for classification? [**2 credits**]

All the same for object. Check which one has the furthest $p(x_i = 1 | \text{Pet})$ from 0,5. ($x_1 = 0,33$; $x_2 = 0,77$; $x_3 = 0,55$)

Check for total error

x_1: 6/15 = 0,40

x_2: 5/15 = 0,33

x_3: 6/15 = 0,40

x_2 is best feature.

2. Try to classify the following objects assuming independence of the features. [**2 credits**]

???	x_1	x_2	x_3
	0	0	0
	1	1	1
	0	1	0
	1	0	1

Likelihood counts

	$x_1 = 1$	$x_2 = 1$	$x_3 = 1$
Pet	3/9	7/9	5/9
Obstacle	3/6	3/6	3/6

Obstacle score always $0{,}5^3 \cdot 0{,}4 = 0{,}05$

???	x_1	x_2	x_3	$p(x \mid \text{Pet})$	Pet score ($\cdot 0{,}6$)	O.S.	Decision	$P(\text{Pet} \mid x)$
	0	0	0	$6/9 \cdot 2/9 \cdot 4/9 = 0{,}0658$	0,0395	0,05	P	0,64
	1	1	1	$3/9 \cdot 7/9 \cdot 5/9 = 0{,}1440$	0,0864	0,05	O	0,63
	0	1	0	$6/9 \cdot 7/9 \cdot 4/9 = 0{,}2305$	0,1383	0,05	O	0,74
	1	0	1	$3/9 \cdot 2/9 \cdot 5/9 = 0{,}0247$	0,0247	0,05	P	0,33

3. If we duplicate the sensor readings ($[x_1, x_2, x_3] \rightarrow [x_1, x_2, x_3, x_1, x_2, x_3]$) would the classifications change? Would we become more confident about them? [**1 credit**]

For this example they wouldn't change. Since the likelihoods are counted twice, $p(x \mid C)$ becomes $p(x \mid C)$. We therefore would become more confident even though no new information was added